

CSA1717-ARTIFICIAL INTELLIGENCE PRACTICAL LAB EXPERIMENTS

BIRD CAN FLY OR NOT

Aim:

To write a Prolog program to determine whether a particular bird can fly.

Objective:

To demonstrate logical inference using facts and rules.

Algorithm: Start.

Define birds.

Define birds that cannot fly.

Define the rule that a bird can fly if it is a bird and is not flightless.

Query the required bird. Display whether it can fly. Stop.

Flowchart:

START

| v

Enter Bird

| v

Is it a Bird?

/ \

No Yes

||

Cannot Is it Flightless? Fly / \

Yes No

||

Cannot Can Fly Fly

\ /

STOP

Prolog Program:

% Bird Flying System

bird(eagle). bird(parrot). bird(pigeon). bird(crow).

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bird(ostrich). bird(penguin).
```

```
cannot_fly(ostrich). cannot_fly(penguin).
```

```
can_fly(Bird) :-bird(Bird),
```

```
\+ cannot_fly(Bird).
```

Output:

Result:

Thus, the Prolog program successfully determines whether a given bird can fly.